



Pezzottaite

△ **Unusual crystal** of pezzottaite showing a trapezoidal structure

Pezzottaite was only formally recognized as a new mineral in 2003, having previously been considered a variety of red beryl (see pp.236–41). Although similar to the crystals found in Utah, USA, its chemical elements differ, and unlike beryl, it crystallizes in the trigonal crystal system. Nonetheless, it has been marketed as raspberyl or raspberry beryl, and it ranges in colour from raspberry red through to orange-red and pink. Most pezzottaite gems are small, between 1 and 2 carats. About 10 per cent will show chatoyancy, or “cat’s eye”.

Specification

Chemical name Cesium, lithium, beryllium silicate
Formula $\text{Cs}(\text{Be}_2\text{Li})\text{Al}_2\text{Si}_6\text{O}_{18}$ | **Colours** Raspberry red, orange-red, pink | **Structure** Hexagonal, trigonal
Hardness 8 | **SG** 3.10 | **RI** 1.601–1.620 | **Lustre** Vitreous
Streak White | **Locations** Afghanistan, Madagascar



Pezzottaite rough | Rough | This fine, gem-quality, 8.40-carat pezzottaite crystal from a pegmatite found at Ambatovy, Madagascar, shows hexagonal form.



Pezzottaite crystal | Rough | Displaying a distinctive form, this pezzottaite crystal is known as an “hourglass” crystal. It features an attractive raspberry tint.



Emerald cut | Cut | A fine-quality, pinkish-lavender pezzottaite weighing 0.71 carats, this stone features an emerald cut. Due to pezzottaite’s rarity, cut stones are usually small.

New discoveries

One mineral, many names

Pezzottaite is one of many recent gem and mineral discoveries. These are often new types of an already well-known gemstone, such as the transparent blue variety of zoisite (known as tanzanite – see p.253), to the discovery of entirely new minerals such as pezzottaite. Other discoveries have revealed previously unknown variations in existing materials. For example, gem tourmaline is thought to be 11 different minerals, all of which are still referred to as tourmaline in the gem trade.



Rutile needles An inclusion deep within this rose-coloured pezzottaite crystal is a spray of rutile needles.



Cat’s eye | Colour variety | This richly coloured Madagascan cabochon weighing 3.46 carats displays superb chatoyancy – a cat’s eye effect created by internal fibres.



Madagascan pezzottaite | Cut | This 4.15-carat oval-cut gem is from Ambatovy in Fianarantsoa Province, Madagascar, one of the few locations where pezzottaite is mined.